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BS Psychology

6th Semester

Assignment Topics

- 1. Sleep: Circadian Rhythm**
- 2. Eating Behavior**
- 3. Weight glucose and Metabolic Regulation**

Sleep: Circadian Rhythm

Physiological and Behavioural description of Sleep

What is Sleep?

Sleep is a behavior and an altered state of consciousness, reduced sensory activity, and inhibited voluntary muscles. It is essential for maintaining overall health and well-being, facilitating processes such as memory consolidation, muscle repair, and the regulation of metabolism and mood.

- Sleep is associated with an urge to lie down for several hours in a quiet environment
- Few movement occur during sleep (eye movements)
- The nature of consciousness is changed during sleep.
- We experience some dreaming during sleep.
- We may recall very little of the mental activity that occurred during sleep

Measures of Sleep

Polysomnography (PSG): A comprehensive sleep study conducted in a lab that records brain
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Sleep Tracking Devices: Wearable technology or apps that monitor sleep stages, duration, and quality using sensors to detect movement and physiological signals. **Electroencephalography (EEG):** Measures electrical activity in the brain to track sleep stages, typically used in combination with other measures in a sleep study.

Home Sleep Apnea Tests (HSAT): Portable devices that measure breathing patterns, oxygen levels, and heart rate to diagnose sleep apnea and other sleep disorders at home.

Stages of Sleep

Sleep is composed of several stages that can be broadly categorized into two main types:

1. Non-rapid eye movement (NREM) sleep
2. Rapid eye movement (REM) sleep.

Rapid Eye Movement (REM)

Characteristics

Timing: REM sleep first occurs approximately 90 minutes after falling asleep and recurs every 90-120 minutes throughout the night. Each subsequent REM period becomes longer, with the final one lasting up to an hour.

Brain Activity: During REM sleep, the brain is highly active, with brain wave patterns resembling those of wakefulness. This is reflected in mixed-frequency waves, including theta and beta waves, seen on an electroencephalogram (EEG).

Eye Movements: Rapid and jerky movements of the eyes occur in different directions, hence the name Rapid Eye Movement.

Physiological Changes

Heart Rate and Blood Pressure increase and become more variable compared to NREM sleep. Respiratory rate becomes irregular and can vary in depth.

Sexual Arousal: Both males and females may experience increased sexual arousal, indicated by erections in males and increased vaginal blood flow in females.

Non rapid eye movement Non (REM)

Non-rapid eye movement (NREM) sleep is one of the two main types of sleep, the other being rapid eye movement (REM) sleep. NREM sleep is divided into three stages, each progressively deeper:

Stage 1 (N1): This is the lightest stage of sleep, where you transition from wakefulness to sleep. It lasts for a few minutes, and during this stage, you might experience muscle jerks and a sensation of falling.

Stage 2 (N2): This is a deeper stage of sleep where you become less aware of your surroundings, your body temperature drops, and your heart rate slows down. It is characterized by specific brain wave patterns called sleep spindles and K-complexes. **Stage 3 (N3):** Also known as slow-wave sleep (SWS) or deep sleep, this stage is crucial for restorative processes in the body. It is characterized by delta waves, which are the slowest brain waves. During this stage, the body repairs and regenerates tissues, builds bone and muscle, and strengthens the immune system.

Mental Activity in Sleep

During sleep, the brain remains active and engages in various mental activities. These include hypnagogic imagery and microdreams during the transition from wakefulness to sleep. Sleep spindles in Stage 2 NREM sleep contribute to memory consolidation. Overall, the brain continues processing information and supporting cognitive functions even during brief periods of sleep.

Neural Regulation

The neural regulation of sleep is a highly orchestrated process involving intricate interactions among various brain regions and neurotransmitter systems. At the core of this regulation is the **hypothalamus**, housing the suprachiasmatic nucleus (SCN), which functions as the body's internal clock, governing circadian rhythms that dictate the timing of sleep and wakefulness. Throughout the **brainstem**, including the reticular formation, critical mechanisms maintain wakefulness and facilitate transitions between different sleep stages. The **thalamus** acts as a relay station, facilitating sensory information processing during wakefulness and modulating sleep stage transitions.

Neurotransmitters such as serotonin, norepinephrine, dopamine, and acetylcholine play pivotal roles: serotonin and norepinephrine promote wakefulness, while GABA and adenosine contribute to sleep promotion. The hypothalamic hormone hypocretin, or orexin, also regulates wakefulness and plays a crucial role in stabilizing sleep-wake transitions. Together, these neural mechanisms ensure the synchronized regulation of sleep, supporting essential physiological processes and cognitive functions critical for overall health and well-being.

Circadian Rhythm

Circadian Rhythm is derived from the Latin words circa dies meaning 'approximately a day'. Term 'Circadian' was coined by Franz Halberg. Circadian rhythm is a 24-hour cycle that tells our bodies when to sleep and regulates many other physiological processes.

The human circadian system is regulated by endogenous clock genes. There is a master clock found in the brain in an anterior section of the hypothalamus nucleus known as the suprachiasmatic nucleus and these are located in the eye, brain, heart, lung, gastrointestinal tract, liver, kidney and fibroblasts.

Sleep Disorders

One common disorder is **insomnia**, characterized by difficulty falling asleep, staying asleep, or experiencing non-restorative sleep despite adequate opportunity.

Another prevalent disorder is **sleep apnea**, where breathing repeatedly stops and starts during sleep, often leading to fragmented sleep and daytime fatigue.

Narcolepsy involves excessive daytime sleepiness and sudden sleep attacks, often accompanied by cataplexy (sudden loss of muscle tone).

Parasomnias like sleepwalking, night terrors, and REM sleep behavior disorder involve abnormal behaviors during sleep.

Eating Behavior

Eating behaviour, in the context of behavioural neuroscience, refers to the complex set of processes involved in food intake, appetite regulation, and energy balance. It encompasses both the physiological and psychological aspects of how we eat and make food-related choices.

Neural Mechanism

Lateral Hypothalamus vs Ventromedial Hypothalamus

The Lateral hypothalamus, also known as the hunger Centre, 'turns on' eating. When it has been switched on, a person will seek out to find food and consume it. Neuropeptide Y (NPY) is an important neurotransmitter which helps turn on eating. An excess amount of this neurotransmitter can cause a mass amount of overeating even when satiated. Damage to the lateral hypothalamus results in a condition called aphagia (lack of eating) – a refusal to eat food even when hungry.

The **Ventromedial hypothalamus**, also known as the satiety centre, does the opposite job, it 'turns off' eating. It sends messages to the brain to tell us that we are full and need to stop eating. But the process is slow which is why we often overeat. Damage to the ventromedial hypothalamus causes a condition of hyperphagia (over eating).

Evidence

Ballistae found that when they damaged the Ventromedial Hypothalamus in rats by making lesions in them, they became obese.

Amygdala & Inferior frontal cortex

The amygdala is the selection of foods based on a previous experience. For example, a person might avoid a certain food due to previously having a bad experience with it. This also means that a person is more likely to select a familiar food over an unfamiliar food because they already know it tastes good.

The **inferior frontal cortex** deals with smells. The olfactory bulb is the part of the brain which is responsible for smell. If a food smells bad or not what we are used to, we would be less likely to eat it because we associate smell with taste. Likewise, if we smell something tasty, we are more inclined to eat it as it appeals to our smell and henceforth, it appeals to our taste.

Serotonin and Dopamine

Serotonin, a neurotransmitter, is involved in the regulation of appetite and mood. Alterations in serotonin signaling have been implicated in conditions such as binge eating disorder and bulimia nervosa.

Dopamine, a neurotransmitter associated with reward and pleasure also influences eating behaviour. It is involved in the rewarding aspects of food consumption and can contribute to the development of compulsive eating behaviours and food addiction.

Eating Disorder

Anorexia Nervosa (Psychological)

The cognitive explanation of anorexia nervosa states that it is a result of faulty thinking which causes irrational beliefs about food. Serotonin and dopamine have been found to contribute to anorexia. Often sufferers of anorexia have misperceptions about their body e.g.- view their body as overweight when its actually underweight. Another form of speaking they experience is the magical thinking. “If I reach a size 6 I will be perfect. Or overgeneralizing thinking “If I fail at controlling my eating I will fail at life”

Evidence McKenzie Anorexic patients judged their ideal weight as lower than normal patients.

Bulimia Nervosa

Bulimia nervosa is an eating disorder characterized by binge eating followed by behaviors to compensate for overeating, such as self-induced vomiting, excessive exercise, or the misuse of laxatives or diuretics. It often involves feelings of guilt, shame, and a distorted body image.

Binge Eating Disorder

Binge Eating Disorder (BED) is a serious eating disorder characterized by recurrent episodes of consuming large quantities of food within a discrete period, typically accompanied by feelings of loss of control and distress. These episodes often occur in secret and are not followed by purging behaviors such as vomiting or excessive exercise, distinguishing BED from bulimia nervosa. Individuals with BED may eat rapidly, beyond the point of feeling full, and often feel ashamed or guilty afterward. The causes of BED are multifactorial, involving biological, psychological, and environmental factors.

Weight, glucose and Metabolic Regulation

Weight, glucose, and metabolic regulation are important aspects of behavioural neuroscience that involve the interplay between the brain, behaviour, and physiological processes related to energy balance.

Weight Regulation: Weight regulation refers to the, processes involved in maintaining a stable body weight. It is influenced by a complex interaction of genetic, environmental, and behavioural factors.

Brain Areas Involved in weight Regulation

Hypothalamus

The hypothalamus plays a central role in weight regulation by integrating various signals to maintain energy homeostasis. The hypothalamus also receives hormonal inputs from leptin, ghrelin, insulin, and other hormones, allowing it to adjust food intake and metabolic rate accordingly.

Brainstem

The brainstem, specifically the nucleus of the solitary tract, receives signals from the gastrointestinal system and conveys information about nutrient availability to the hypothalamus. This helps coordinate the regulation of food intake and energy expenditure.

Neurotransmitters

Neuropeptide Y (NPY):Produced in the arcuate nucleus (ARC) of the hypothalamus. Strongly stimulates appetite and food intake. NPY levels increase during fasting and decrease after eating.

Agouti-Related Peptide (AgRP):Co-expressed with NPY in the ARC.Increases appetite by antagonizing melanocortin receptors.

Dopamine: Produced in the ventral tegmental area (VTA) and acts on the nucleus accumbens. Involved in the reward and pleasure aspects of eating. Dopamine levels increase in response to palatable food, reinforcing eating behavior.

Serotonin (5-HT):Modulates appetite and mood. Increased serotonin activity generally suppresses appetite. Certain serotonin receptors, like 5-HT_{2C}, are known to reduce food intake.

Glucose regulation

Glucose regulation is a critical process in the body that ensures blood glucose levels remain within a narrow range, providing a steady supply of energy to cells while preventing hyperglycemia (high blood glucose) or hypoglycemia (low blood glucose).

Hormones

Insulin: Produced by beta cells in the pancreas. Lowers blood glucose levels by promoting the uptake of glucose into cells, particularly in the liver, muscle, and adipose tissue.

Glucagon: Produced by alpha cells in the pancreas. Raises blood glucose levels by stimulating glycogenolysis and gluconeogenesis in the liver. Works antagonistically to insulin.

Epinephrine (Adrenaline): Produced by the adrenal medulla. Raises blood glucose levels by stimulating glycogenolysis and gluconeogenesis. Inhibits insulin secretion and stimulates glucagon release.

Cortisol: Produced by the adrenal cortex. Increases blood glucose levels by promoting gluconeogenesis and reducing the uptake of glucose by cells.

Brain Areas Involved in Glucose regulation

- The regulation of glucose involves several brain areas, primarily within the **hypothalamus**, which plays a pivotal role in maintaining glucose homeostasis. The hypothalamus monitors blood glucose levels and integrates hormonal signals to coordinate appropriate responses.
- The **autonomic nervous system**, including sympathetic and parasympathetic pathways, further modulates glucose regulation by affecting insulin and glucagon secretion from the pancreas and glucose production and uptake in peripheral tissues.

Metabolic Regulation

Metabolic regulation refers to the complex mechanisms that control the body's metabolic pathways, ensuring that energy production, storage, and expenditure are balanced to meet the physiological needs of the organism.

Hormones

Insulin: Produced by the pancreas. Facilitates glucose uptake by cells, promotes glycogenesis (storage of glucose as glycogen), and inhibits gluconeogenesis (production of glucose from non-carbohydrate sources).

Cortisol: Produced by the adrenal cortex. Increases gluconeogenesis, promotes the breakdown of proteins and fats, and inhibits glucose uptake in muscle and adipose tissue.

Leptin: Produced by adipose tissue. Signals the hypothalamus to suppress appetite and increase energy expenditure when fat stores are sufficient.

Ghrelin: Produced by the stomach. Stimulates hunger by acting on the hypothalamus, increasing food intake.

Brain Areas Involved in Metabolic Regulation

- The hypothalamus regulates energy expenditure through the sympathetic nervous system, which controls functions such as thermogenesis (heat production) and the breakdown of stored energy sources like fat.

- The **brainstem** integrates signals from the gastrointestinal system, liver, and adipose tissue to regulate metabolism. It, can influence food intake, nutrient absorption, and metabolic responses to changes in energy availability.

